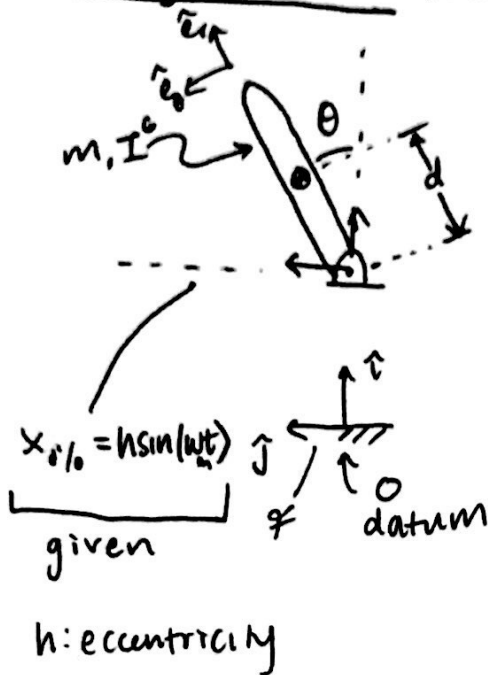


Today:

- 1) Shaking Pendulum (Lagrange)
- 2) Double pendulum (1 of 3)

Shaking Pendulum: (same pic as before)



Lagrange equations

• Minimum coordinate:  $\theta$

want in terms of  $\theta, \dot{\theta}$

$$\begin{cases} E_k = \frac{1}{2} m \vec{v}_G^2 + \frac{1}{2} I_G \dot{\theta}^2 \\ E_p = mg (\text{distance up}) \end{cases}$$

$\dot{\theta}, \neq w_m$

$$E_k = \frac{1}{2} m \vec{v}_G \cdot \vec{v}_G + \frac{1}{2} I_G \dot{\theta}^2$$

$$\vec{v}_G = \vec{v}_{o/o} + \vec{v}_{G/o}$$

$$\vec{v}_{o/o} = \frac{d}{dt} (y_{o/o}) = h w_m \cos(w_m t)$$

$$\vec{v}_{G/o} = d \dot{\theta} \hat{e}_\theta$$

$$= \text{mess}(\theta, \dot{\theta}, \sin, \cos)$$

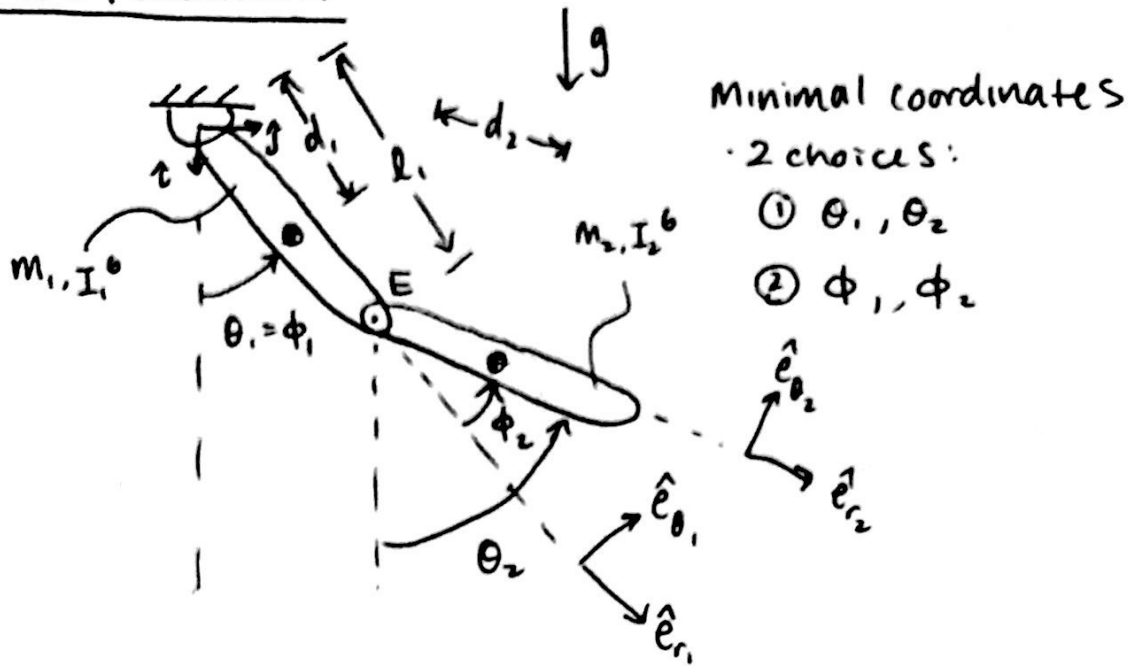
$$E_p = (h \sin(w_m t) + d \cos \theta) mg$$

$$\mathcal{L}(\theta, \dot{\theta}) = E_k - E_p$$

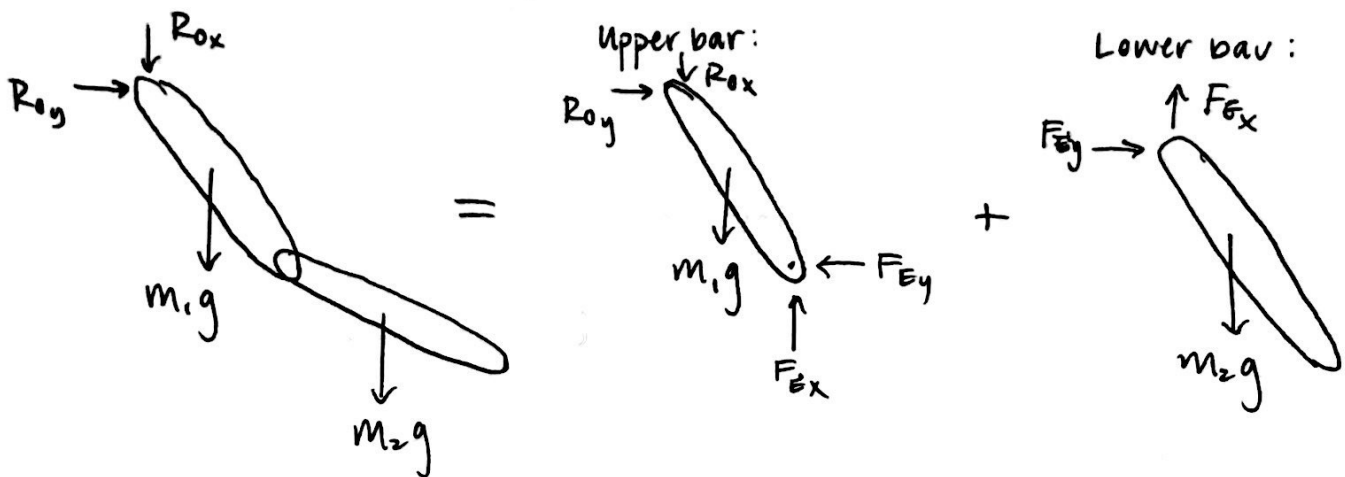
L.E.:  $\frac{d}{dt} \left( \frac{\partial \mathcal{L}}{\partial \dot{\theta}} \right) - \frac{\partial \mathcal{L}}{\partial \theta} = 0$

one eqn no non conservative forces

# Double Pendulum:



3 FBDS:



$\theta_1 + \theta_2$  vs.  $\phi_1 + \phi_2$   
 conceptually easier  
 more standard  
 harder  
 good for robotics (motor acts on  $\phi$ , not  $\theta$ )

N-E:

1)  $\text{AMB}_{/0}$  for system

$$\sum \vec{M}_{/0}^{\text{ext}} = \dot{\vec{H}}_{/0}$$

2)  $\text{AMB}_{/E}$  for link 2

$$\sum \vec{M}_{/E}^{\text{ext}} = \dot{\vec{H}}_{/E}$$

note: E is non-inertial pt.  
(have to use acceleration form of eqns)